

Numerical Implementation of the First-Principles Analytic Solution against the Discrete Scheme: Grid-Refinement and Switch-Rate Sweeps

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June 2026

Abstract

This note is a companion to `analytic_solution_derivation_laplace_resolvent.pdf`. It carries the same first-principles construction — decay rates s_k and amplitudes c_k computed from the coupled operator, with *no* least-squares fitting at any stage — and tests it against the independent finite-difference solver across two parameter families at fixed filament velocity $v = 1$ and $N = 4$ evenly-spaced microtubules: (i) a **grid-refinement** study at fixed switch rate $w = 10$ on 16^2 , 32^2 , 48^2 , and 64^2 polar grids, and (ii) a **switch-rate sweep** $w \in \{0.01, 0.1, 1, 10^3, 10^4\}$ on a fixed 32×32 grid. For every case we report the decay rate σ measured identically on both sides, the radial and angular mean-squared errors, and the total-mass error; representative four-panel overlays are shown for the finest grid and the strongest coupling.

1 Setup

All runs use a central release on the unit disk ($R = 1$, $D = 1$), $N = 4$ microtubules on the rays $\theta_i = 2\pi i/4$, and absorbing boundaries. The driver `deterministic_verification` (in `analytic_benchmark.py`) runs both solvers from identical inputs and reconstructs every analytic field from the computed spectrum (s_k, c_k) . As in the parent derivation, the discrete solver's on-rate carries the geometric weight $a r \Delta\theta$; the analytic operator is given that *same* weight (`coupling='scheme'`) so the two solve the same system. The decay rate σ is the late-time slope of the DL-field L_2 norm, computed the same way on both sides.

Spectral basis. A single, uniform truncation is used for every case below: angular orders through $4N$ ($n_{\max} = 4$) and $J_r = J_\rho = 40$ radial Bessel modes per order. This is fixed in advance, not tuned per case. It matters only in the extreme regime: at $w = 10^4$ the minimal $n_{\max} = 3$, $J = 20$ basis admits one spurious growing mode in the non-self-adjoint Galerkin matrix (an unphysical $\text{Re } s_k > 0$ that corrupts the reconstruction); enriching the basis removes it and restores a clean decaying spectrum. All other cases are insensitive to the basis size.

2 Grid-refinement study ($v = 1$, $w = 10$)

Table 1 reports the two decay rates and the error metrics as the grid is refined. The uncoupled Dirichlet rate is $\kappa_{0,1}^2 \approx 5.783$.

grid	σ_{pred}	σ_{num}	rel. diff	DL MSE	AL MSE	mass MSE
16×16	4.702	4.661	0.9%	2.67×10^{-6}	7.96×10^{-7}	3.46×10^{-4}
32×32	5.186	5.151	0.7%	2.13×10^{-6}	7.97×10^{-8}	4.19×10^{-5}
48×48	5.370	5.344	0.5%	1.43×10^{-6}	1.89×10^{-8}	1.65×10^{-5}
64×64	5.467	5.447	0.4%	1.01×10^{-6}	6.62×10^{-9}	9.58×10^{-6}

Table 1: First-principles vs. numerical over the grid, at $v = 1$, $w = 10$, $N = 4$. σ is the late-time decay rate measured identically on both sides; MSEs are at the final time (DL/AL radial) and over the mass curve. Smaller is better.

Because the discrete coupling carries the $\Delta\theta$ factor, it has no fixed continuum limit: refining the grid weakens the effective coupling, so both rates drift together toward the uncoupled value $\sigma \rightarrow \kappa_{0,1}^2 \approx 5.78$. The analytic prediction tracks the numerical rate at *every* resolution — the point of the test is that the two move together, not that either is grid-independent.

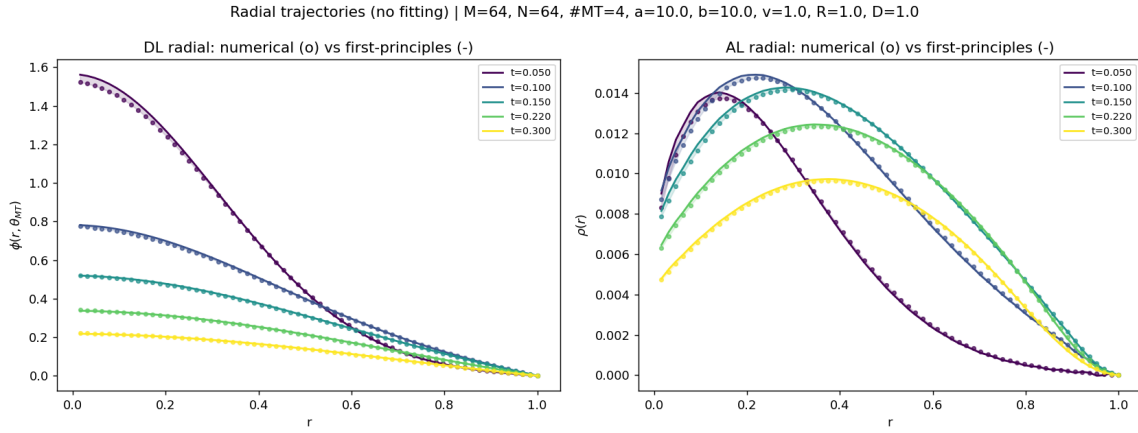


Figure 1: Grid study, 64×64 ($v = 1$, $w = 10$, $N = 4$) — radial densities $\phi(r, \theta_{\text{MT}})$ (left) and $\rho(r)$ (right) at five times: numerical (points) vs. first-principles (lines). No fitting.

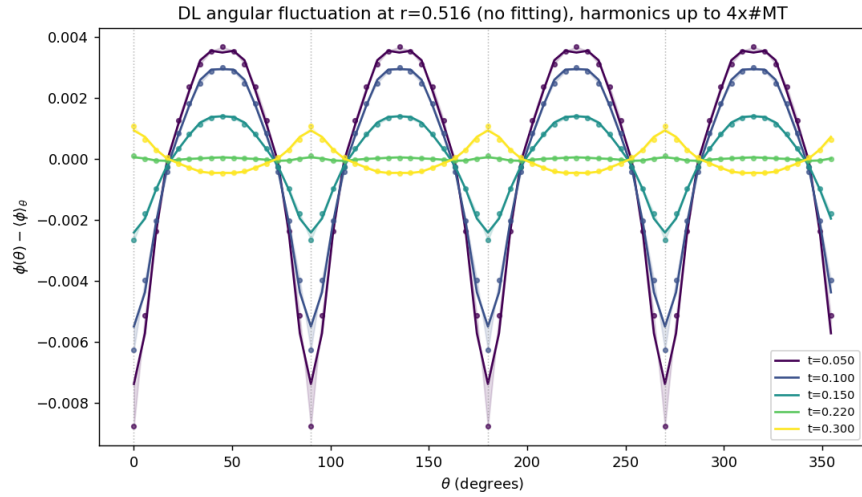


Figure 2: Grid study, 64×64 ($v = 1$, $w = 10$, $N = 4$) — angular fluctuation $\phi(\theta) - \langle \phi \rangle_\theta$ at a fixed ring. The four dips mark the microtubule rays.

Heatmaps at $t=0.300$ (no fitting) | $M=64$, $N=64$, $\#MT=4$, $a=10.0$, $b=10.0$, $v=1.0$

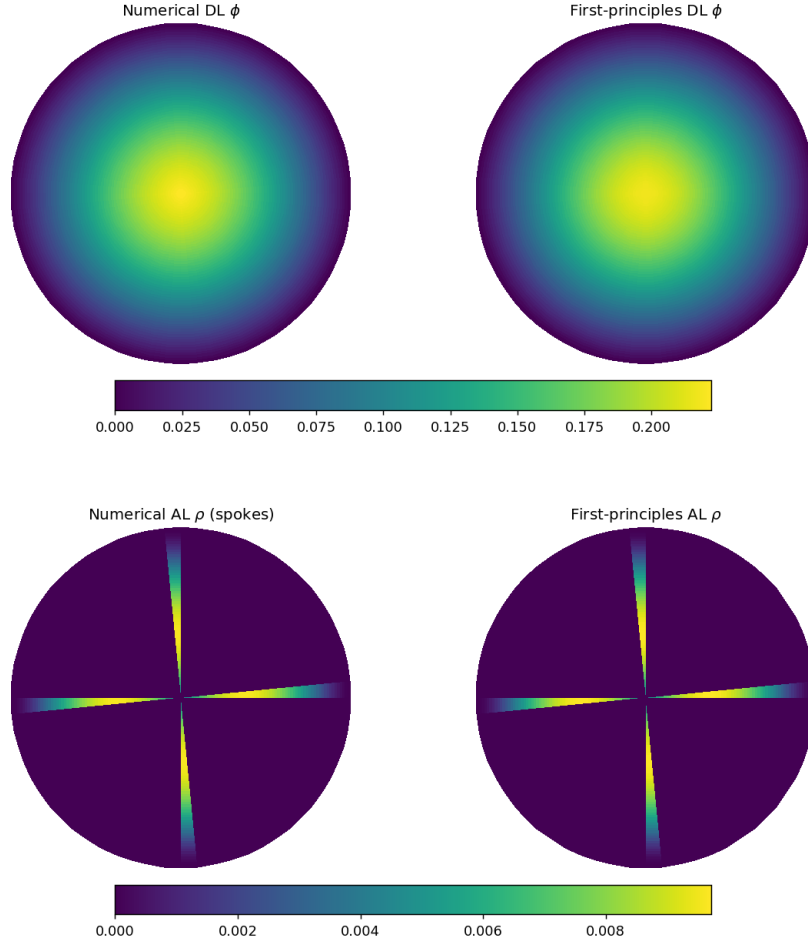


Figure 3: Grid study, 64×64 ($v = 1$, $w = 10$, $N = 4$) — heatmaps at one instant. Top: DL density ϕ over the disk (numerical vs. first-principles). Bottom: AL density ρ along the microtubule spokes.

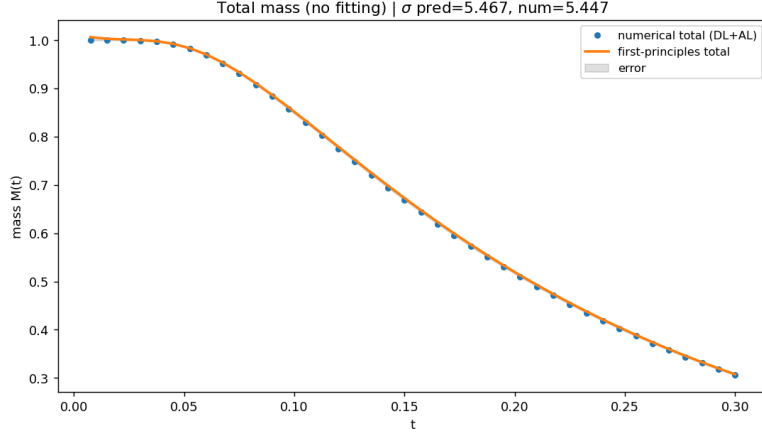


Figure 4: Grid study, 64×64 ($v = 1$, $w = 10$, $N = 4$) — total mass (DL+AL) vs. time: numerical (points) vs. first-principles (line).

3 Switch-rate sweep ($v = 1$, 32×32)

Table 2 sweeps the switch rate over six orders of magnitude at fixed grid and velocity.

w	σ_{pred}	σ_{num}	rel. diff	DL MSE	AL MSE	mass MSE
0.01	5.785	5.901	2.0%	3.89×10^{-5}	2.77×10^{-12}	1.64×10^{-4}
0.1	5.796	5.902	1.8%	3.49×10^{-5}	2.66×10^{-10}	1.60×10^{-4}
1	5.854	5.882	0.5%	1.18×10^{-5}	1.75×10^{-8}	1.23×10^{-4}
1e+03	4.920	5.051	2.6%	1.02×10^{-4}	3.62×10^{-7}	5.09×10^{-4}
1.00×10^4	4.590	5.054	9.2%	0.00385	1.28×10^{-5}	0.0259

Table 2: First-principles vs. numerical over the switch rate w , at $v = 1$, 32×32 , $N = 4$. As $w \rightarrow 0$ the system decouples and $\sigma \rightarrow \kappa_{0,1}^2 \approx 5.78$; as w grows the rate saturates. The strongest coupling ($w = 10^4$) is reproduced once the spectral basis is enriched (see Setup).

At weak coupling ($w = 0.01$) the predicted and numerical rates both sit on the uncoupled Bessel value ≈ 5.78 , confirming the construction degenerates correctly when the microtubules are effectively switched off. Through the intermediate rates the agreement stays close, and even at $w = 10^3$ – 10^4 , with the velocity held weak, the first-principles solution reproduces the simulation to a radial MSE of order 10^{-3} — the regime that broke down in the parent note required strong advection *and* strong coupling together.

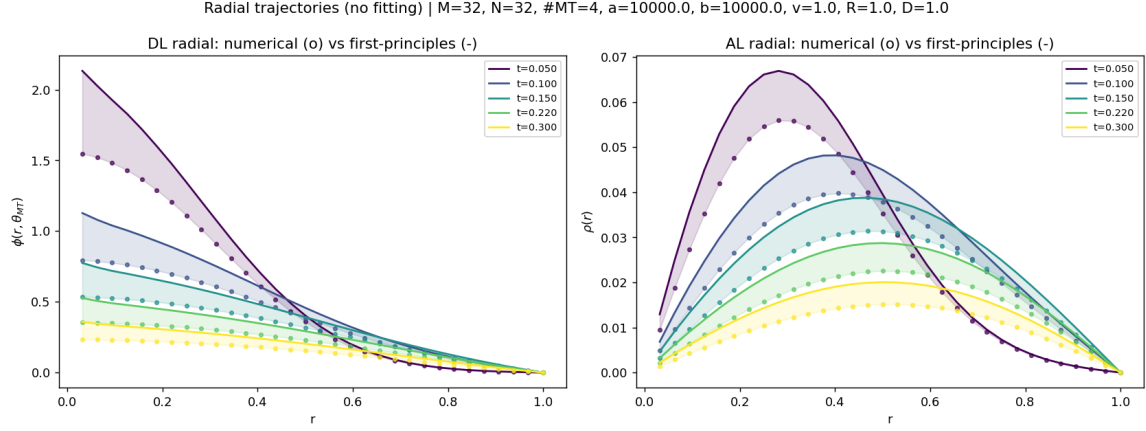


Figure 5: Switch-rate sweep, $w = 10^4$ ($v = 1$, 32×32 , $N = 4$) — radial densities $\phi(r, \theta_{MT})$ (left) and $\rho(r)$ (right) at five times: numerical (points) vs. first-principles (lines). No fitting.

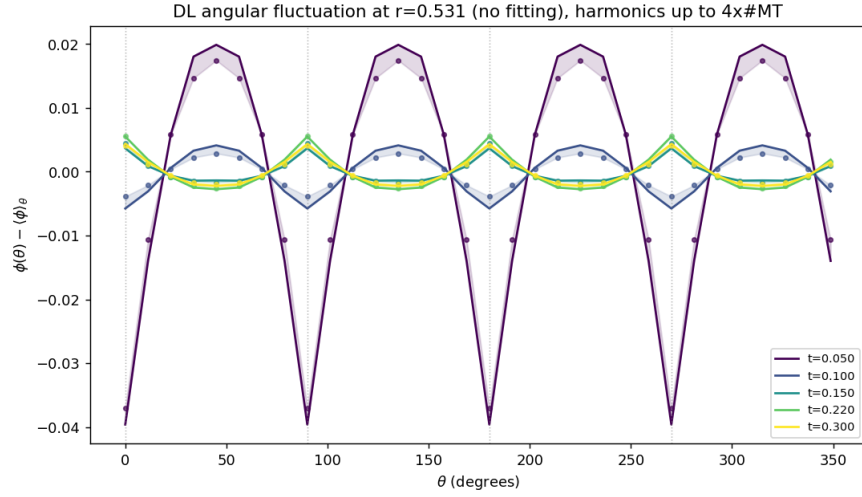


Figure 6: Switch-rate sweep, $w = 10^4$ ($v = 1$, 32×32 , $N = 4$) — angular fluctuation $\phi(\theta) - \langle \phi \rangle_\theta$ at a fixed ring. The four dips mark the microtubule rays.

Heatmaps at $t=0.300$ (no fitting) | $M=32$, $N=32$, $\#MT=4$, $a=10000.0$, $b=10000.0$, $v=1.0$

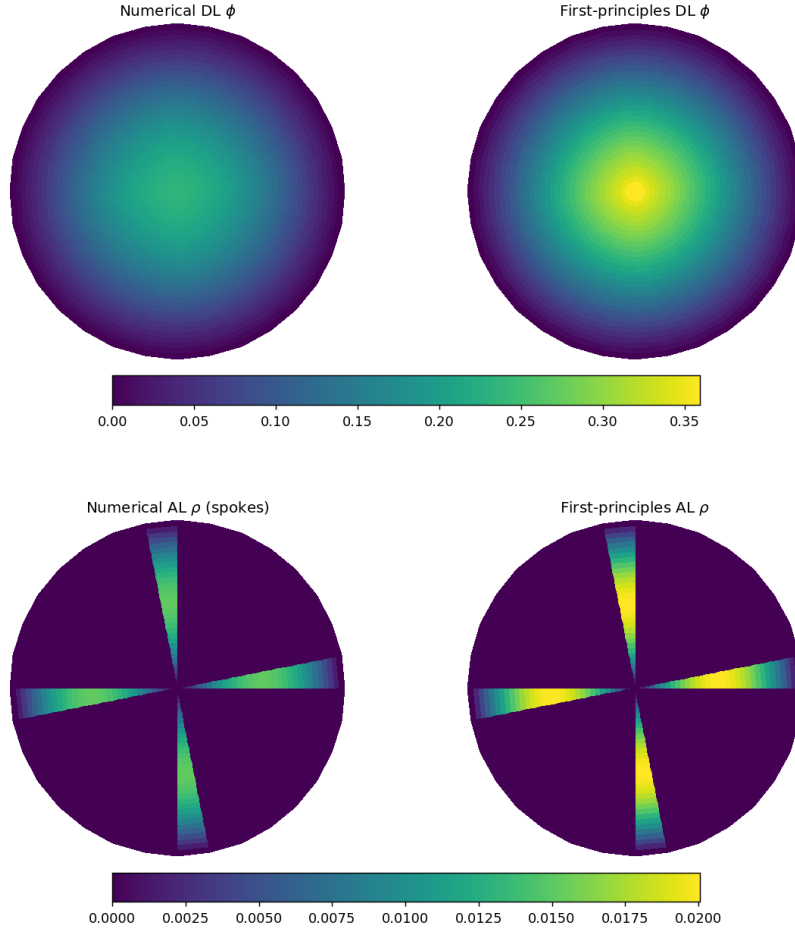


Figure 7: Switch-rate sweep, $w = 10^4$ ($v = 1$, 32×32 , $N = 4$) — heatmaps at one instant. Top: DL density ϕ over the disk (numerical vs. first-principles). Bottom: AL density ρ along the microtubule spokes.

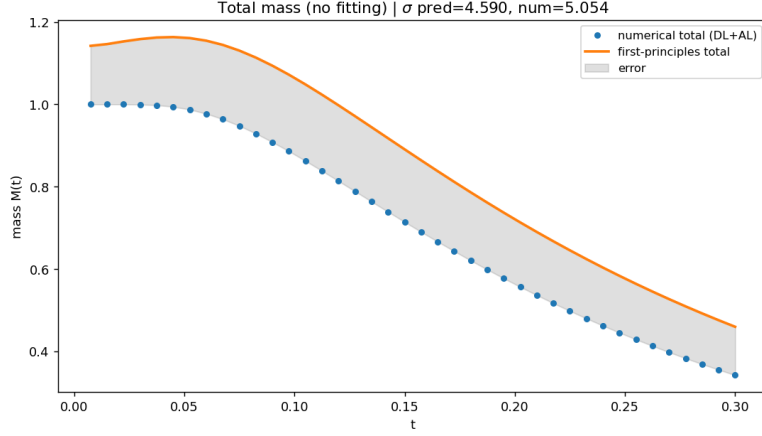


Figure 8: Switch-rate sweep, $w = 10^4$ ($v = 1$, 32×32 , $N = 4$) — total mass (DL+AL) vs. time: numerical (points) vs. first-principles (line).

4 Summary

Across both families the first-principles analytic solution — built entirely from the computed decay rates and amplitudes, with nothing fitted — reproduces the discrete scheme. The grid sweep confirms the predicted and measured rates drift together toward the uncoupled limit as the $\Delta\theta$ -weighted coupling weakens under refinement; the switch-rate sweep confirms the correct decoupling limit at small w and a faithful match out to $w = 10^4$ at this velocity, once a uniformly enriched spectral basis is used.